
LAB EXPERIMENTS 27 28 29

NAME: J SUBHAASHREE

REG NO: 192424008

EXPERIMENT 27

Aim

To create a vertical bar chart showing the popularity of different programming languages using Python, Pandas, and Matplotlib.

Dataset Description

```
Language,Popularity
Java,22.2
Python,17.6
PHP,8.8
JavaScript,8.0
C#,7.7
C++,6.7
```

Procedure

1. Import the Pandas and Matplotlib libraries.
2. Read the dataset from the CSV file.
3. Store the language names and popularity values.
4. Create a vertical bar chart using the bar() function.
5. Add labels, title, and axis descriptions.
6. Display the chart.

Code

```
exp27.py - C:/Users/Harshini Darshan/OneDrive/Desktop/QUERY_PROCE
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("data/language_popularity.csv")

print(df)

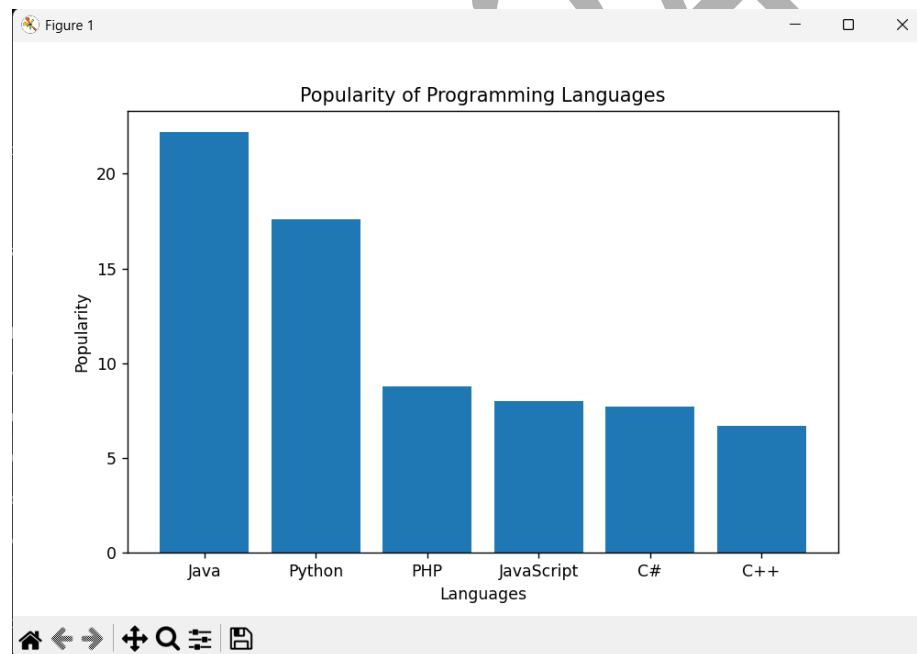
plt.figure(figsize=(8,5))

plt.bar(df["Language"], df["Popularity"])

plt.xlabel("Languages")
plt.ylabel("Popularity")
plt.title("Popularity of Programming Languages")

plt.show()
```

Output



Result

The vertical bar chart was successfully created using the programming language popularity dataset. The chart clearly shows the comparative popularity of different programming languages.

EXPERIMENT 28

Aim

To create a horizontal bar chart showing the popularity of different programming languages using Python, Pandas, and Matplotlib.

Dataset Description

```
Language,Popularity
Java,22.2
Python,17.6
PHP,8.8
JavaScript,8.0
C#,7.7
C++,6.7
```

Procedure

1. Import the required libraries.
2. Load the dataset using Pandas.
3. Extract the language names and popularity values.
4. Create a horizontal bar chart using the `barh()` function.
5. Add suitable labels and title.
6. Display the chart.

Code

```
exp28.py - C:/Users/Harshini Darshan/OneDrive/Desktop/QUERY_PROCESSI
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("data/language_popularity.csv")

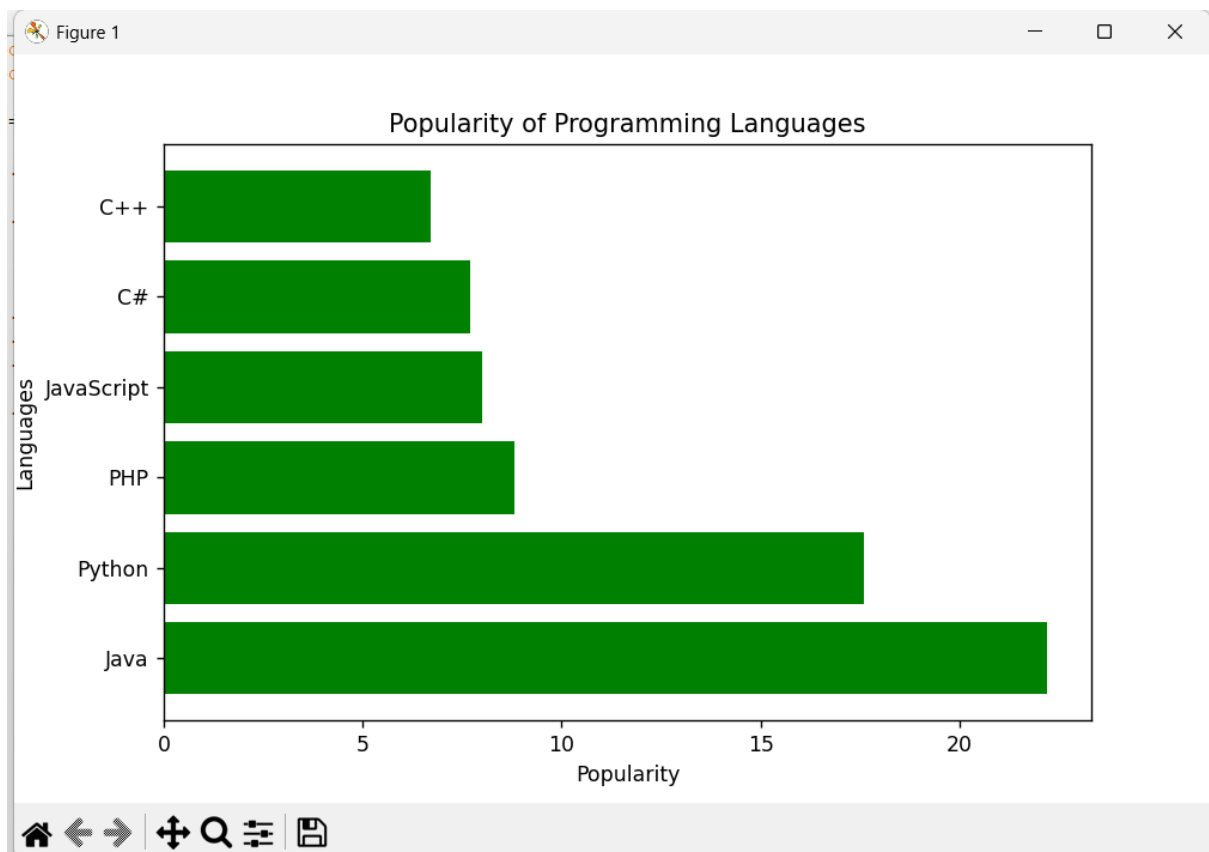
plt.figure(figsize=(8,5))

plt.barh(df["Language"],
         df["Popularity"],
         color="green")

plt.xlabel("Popularity")
plt.ylabel("Languages")
plt.title("Popularity of Programming Languages")

plt.show()
```

Output



Result

The horizontal bar chart was successfully created and displayed. The chart effectively visualizes the popularity of programming languages in a horizontal format.

EXPERIMENT 29

Aim

To create a bar chart showing the popularity of programming languages using different colors for each bar.

Dataset

```
Language,Popularity
Java,22.2
Python,17.6
PHP,8.8
JavaScript,8.0
C#,7.7
C++,6.7
```

Procedure

1. Import Pandas and Matplotlib libraries.
2. Read the dataset from the CSV file.
3. Define a different color for each programming language.
4. Create a bar chart using the bar() function.
5. Apply different colors to each bar.
6. Add chart title and axis labels.
7. Display the chart.

Code

```
exp29.py - C:/Users/Harshini Darshan/OneDrive/Desktop/QUERY_PROCES
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("data/language_popularity.csv")

colors = ['red',
          'black',
          'green',
          'blue',
          'yellow',
          'cyan']

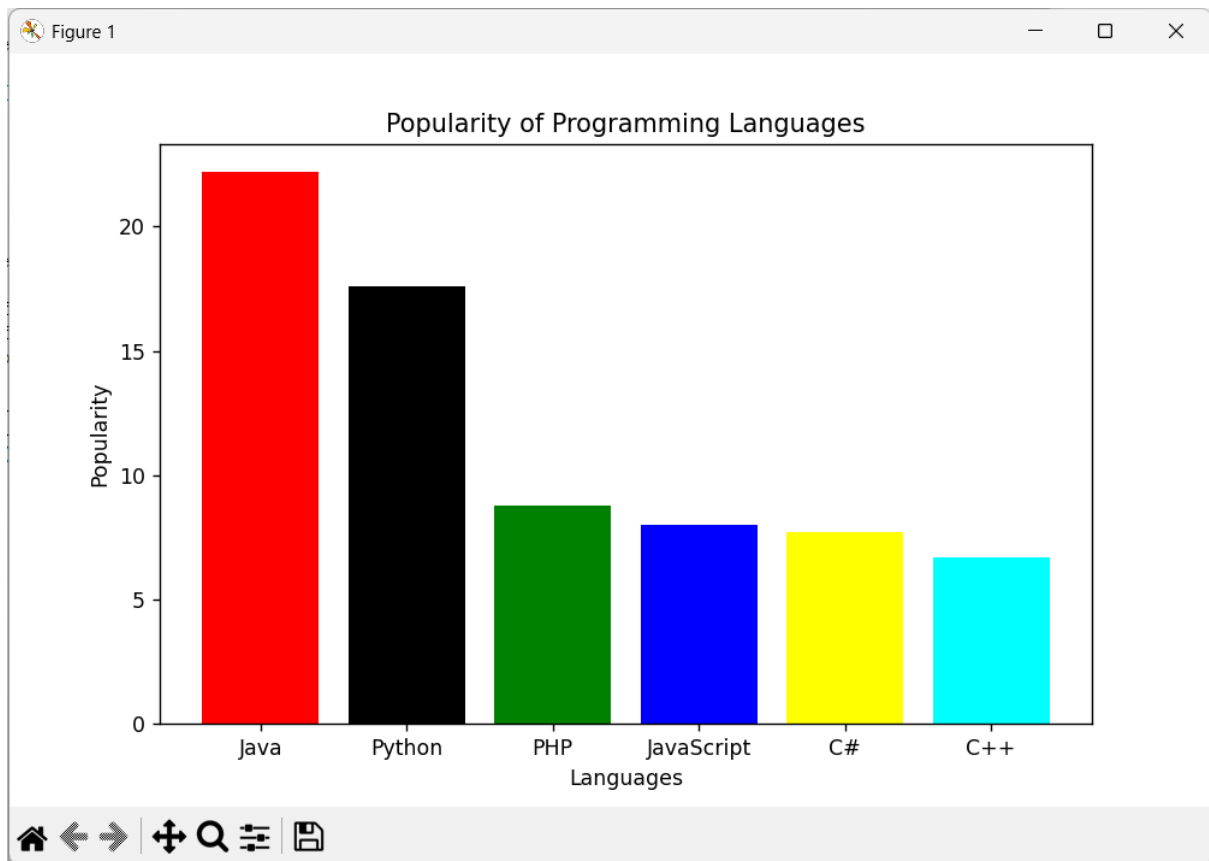
plt.figure(figsize=(8,5))

plt.bar(df["Language"],
        df["Popularity"],
        color=colors)

plt.xlabel("Languages")
plt.ylabel("Popularity")
plt.title("Popularity of Programming Languages")

plt.show()
```

Output



Result

The colored bar chart was successfully generated using Matplotlib. Different colors were assigned to each programming language, making the chart more attractive and easier to interpret.